

# Questions to be covered :

- Scan Insertion
  - What is Scan insertion?
  - Need of Scan Insertion
  - Scan Cell, Scan chain
  - Scan chain operation
  - Flow of Scan insertion in Tessent tool
  - Inputs and Outputs of Scan insertion
  - Why scan insertion is mostly performed on a gate level netlist?
  - Scan DRCs and its fixes
  - Different ways to fix the DRCs in Tessent tool
  - Need of balancing
  - Edge mixing and Domain mixing
  - Problems in edge and domain mixing and its fixes

# Questions to be covered (Contd.):

- Scan Compression
  - What is scan compression?
  - What is the need of scan compression?
    - Test time and Test data volume reduction. (Logical explanation & Using formulae)
  - Compression ratio
  - Merits and De-merits of high and low compression ratio
  - EDT logic
    - Decompressor
    - Compressor
    - Controller (Mask shift register, Mask hold register, Decoders)
    - Bypass logic
  - Tessent TestKompress Flow for EDT insertion
  - Inputs and Outputs of EDT insertion

# Questions to be covered (Contd...):

- Automatic Test Pattern Generation
  - What is ATPG?
  - Defect, fault and error
  - Fault simulation
  - Questions related to finding out the pattern which will detect a particular fault
  - Fault models (Stuck at, TDF, PDF, IDDQ)
  - Dead cycles
  - Pipelined Scan En
  - Fault categories (Fault dominance, Fault equivalence, Fault collapse)
  - Fault aliasing
  - Fault classes (Testable, Untestable faults and their sub-classes)
  - Coverage, Test Coverage, Fault Coverage
  - Pattern classification
  - Tessent TestKompress flow of ATPG
  - Inputs and Outputs of ATPG
  - Coverage Analysis and Improvement
  - Why TDF coverage is less than SA coverage?
  - Test Point Insertions
  - Bypass patterns

## ATPG :

1. Fault Sim

2. Coverage

imp

→ PC

→ SEQ

→ UO, UL  
(TPI)

→ mem

→ EAS

3. SEQ depth  
concept

4. TDF ; str and  
sh path  
gm ;

waveform ;  
coc, cos and comp

In ATPG,  
these are the must do topics.

# Questions to be covered (Contd..):

- On-chip clock controller
  - Need of OCC
  - OCC Architecture
  - Placement of OCC
- Simulations (Only no-timing simulations)
  - Need of simulations
  - Flow of simulations (using Questa Sim tool)
  - Simulation mismatch debug

# Questions to be covered (Contd...):

- Joint Test Action Group (JTAG IEEE 1149.1 standard)
  - Need of JTAG
  - JTAG Tap Architecture
  - JTAG FSM
  - How static signals are controlled using JTAG?
  - What is Boundary Scan and its need?
  - Boundary Scan cell
  - Modes of operations of Boundary scan cell
  - Extest, Sample/Pre-load, bypass instructions

# Questions to be covered (Contd...):

- IEEE 1687 Standard IJTAG
  - Architecture
  - Comparison between JTAG and IJTAG

- Memory Built-in Self Test (MBIST)

- Memory Fault models
- Memory Algorithms *← if time permits → algo seq + which algo will be able to cover which faults in mem.*
- Tessent MBIST Implementation and Architecture
- Tessent MBIST operation protocol
- Tessent MBIST flow
- Inputs and Outputs of MBIST insertion
- Group and Step Concept
- MBIST pattern specifications and simulations
- MBIST simulation mismatch debug

*not v. imp* ←

# Questions to be covered (Contd...):

- Hierarchical Scan
  - What is Hierarchical Scan?
  - TSDB flow
  - MBIST and IJTAG insertion (in RTL) (understanding inputs and outputs, Tessent flow and DRC cleanup)
  - EDT and OCC insertion (in RTL)
  - Scan Insertion
  - Scan wrapper
  - Graybox generation
  - ATPG (intest, extest runs) and simulations
  - Pattern Re-targeting



# Previous Companies Difficult interview Questions

## **Scan Insertions :**

- Explain Scan chain shifting using waveform
- Why do we need more than 1 scan clock in the design?
- Why scan insertion is performed on gate level netlist?
- Why PD team require scandef file? What will happen if scandef file is not provided to PD team?
- Why do we have non-scan flops in the design?
- What steps we take while doing scan planning?

## **Scan Compression :**

- Why do we have XOR gates inside the compressor of EDT logic?
- If there are un-equal channels in EDT, how to calculate compression ratio? Is there any limitations of having unequal channels in EDT?

# Previous Companies Difficult interview Questions (Contd...)

## ATPG :

- Why do we need slow frequency clock in the capture phase of stuck-at fault model and fast frequency clock in the capture phase of TDF?
- Explain SA and TDF fault model using waveforms
- What is the advantage of declaring reset as a clock?
- What is the advantage of connecting TE of ICG to SE?
- Reset is considered as a clock in order to get coverage during stuck-at pattern generation. But in at-speed generation, reset faults are not covered. Why?
- How to reduce pattern count without affecting coverage?
- If we provide controllability and observability to PIs and POs, will it improve the test coverage and how much percent improvement can we see?
- Test Plan for ATPG

# Previous Companies Difficult interview Questions (Contd...)

## **Simulations :**

- Why do we need no-timing simulations?
- What is unit delay simulations?
- Why do chain pattern simulation fail? During ATPG we have done the tracing checks, then why do we get simulation failures?
- What is binary mismatch and x-mismatch?

## **JTAG :**

- Why TRST is optional in JTAG?
- JTAG FSM is mealy or moore machine?
- What is the default state/value of JTAG pins (TDI, TDO, TMS)